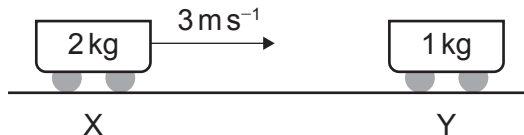


PAPER 1 / QUESTION 75

5 November 2025 / TZ1 (file label) / HL Paper 1A / original question 4 / 8825-6707

75. Cart X, of mass 2 kg, is moving at a speed of 3 m s^{-1} to the right and collides on a horizontal track with cart Y of mass 1 kg. Y is initially stationary.



The velocity of Y immediately after the collision is 4 m s^{-1} to the right. What is the velocity of X immediately after the collision?

- A. 1 m s^{-1} to the right
- B. 1 m s^{-1} to the left
- C. 2 m s^{-1} to the right
- D. 2 m s^{-1} to the left